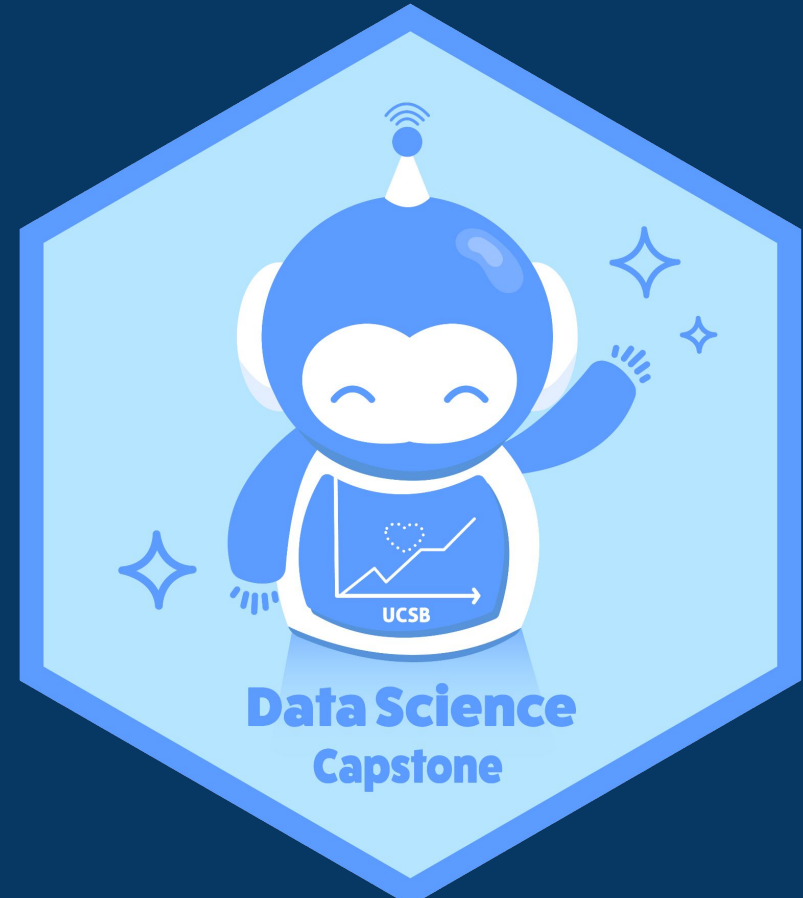


Improving Transparency: Interactive CalCOFI Data Access Portal

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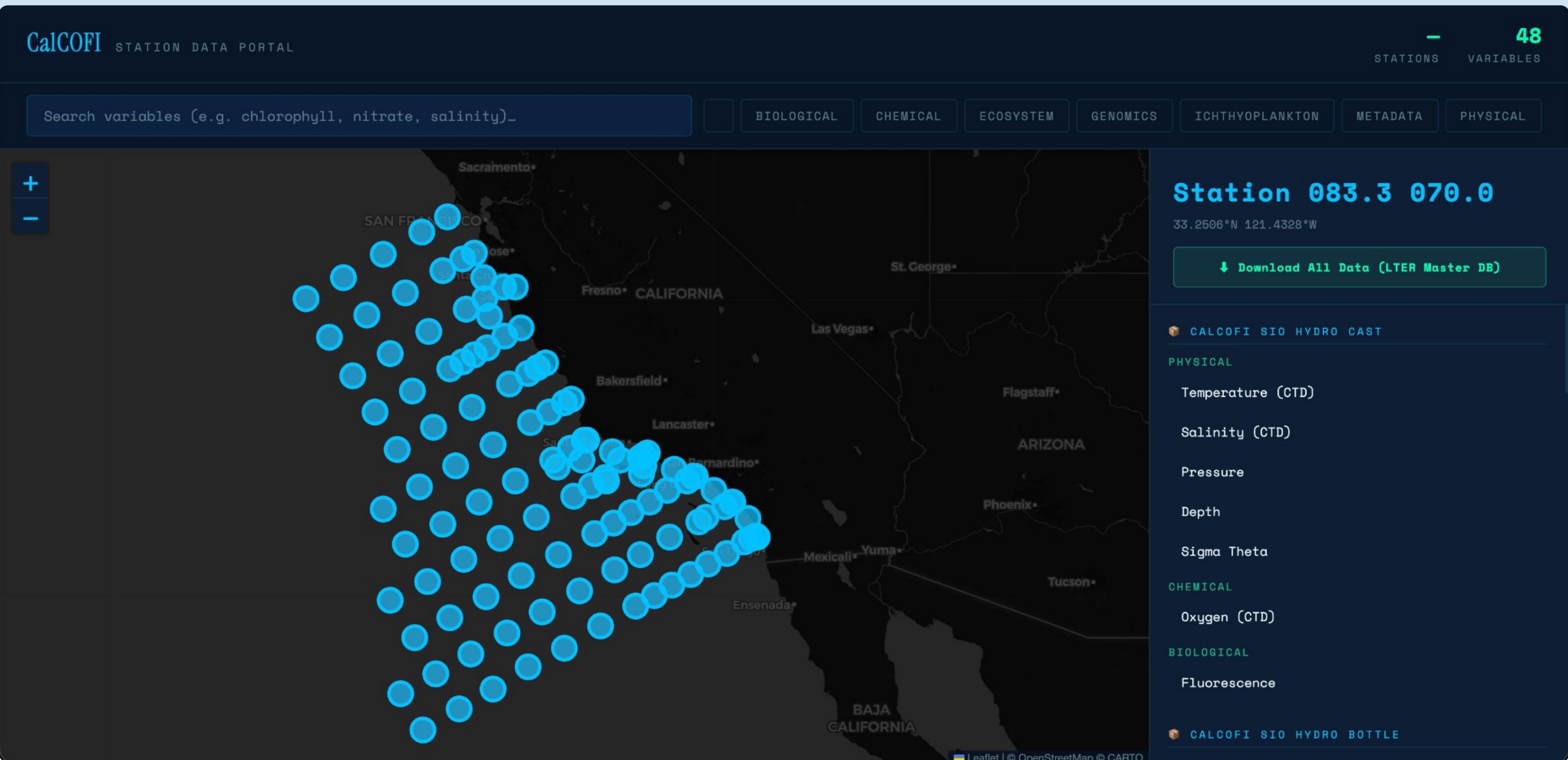
Abstract

Develop a **user-friendly, variable-based CalCOFI data discovery inventory** that helps users quickly find where CalCOFI data reside across multiple repositories and platforms. Since CalCOFI is a distributed system of different data providers sampling on the CalCOFI platform this serves as a federated system. The tool will enable users to browse or **search by variable, parameter type, or dataset** and directly link to the appropriate data source for their need (e.g., across major repositories and data serving services – ERDDAP, OBIS, EDI, NCEI, program databases, and visualization tools). Because CalCOFI data are heterogeneous in format, stewardship, and delivery systems, it is not realistic to centralize all datasets within a single platform. Instead, the inventory will function as a **navigation layer** organized around **standardized variable names** and **dataset descriptions**, with clearly labeled links to landing pages, DOIs, dynamic services, and APIs.

CalCOFI Pain Points

- **Disaggregated Data**
 - CalCOFI data is spread into dozens of different datasets which all carry similar information about collection, but it's dispersed across various modes
- **Transparency Struggle**
 - CalCOFI executives are struggling to find the data about stations, data collection methods, locations, and other aspects about the operation.
- **Delayed Analysis**
 - Due to these factors, data analytics is can become delayed and the rate of progress possibly is slowed.

Final Product



Key Outcomes/Use Cases

- Created a **transparent discovery layer** for distributed CalCOFI data
- **Improved station-level visibility** of what data exist and where they can be found
- Supports **faster onboarding** for students, researchers, and collaborators
- **Reduces time spent** manually searching across multiple repositories
- Provides a **foundation for future data integration** and reproducible workflows

“The CalCOFI team’s work will help not only us at CalCOFI, Scripps, and the Sea Grant, but any researcher or scientist who is looking for data. Their work has dramatically helped reduce the friction that we’ve seen with data collection.”

Future Developments

For the future, CalCOFI and associated groups can use our platform to access data that was previously confusing to access. To extend this project to the future, creating a standardized / centralized data environment would allow for more specific access to data values.

As a more centralized data environment is created, more advanced variable sorting and filtering could be implemented for more efficient use cases.

References and Acknowledgments

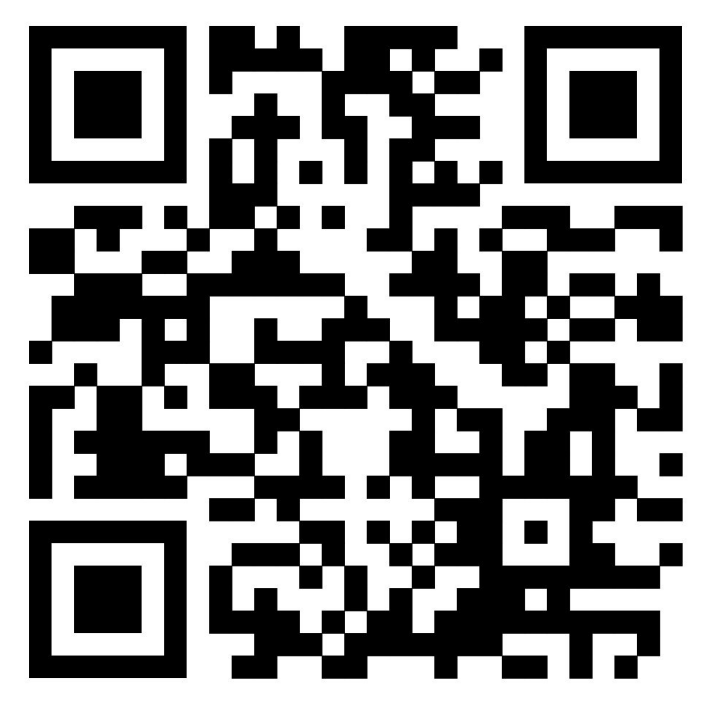
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Project Team: Aashish Krishnan, Nicole Xu, Lucy Cao

Mentor: Dr. Erin Satterthwaite.

UCSB Advisor: Dr. Tony Coburn.

Web App



Demo Video



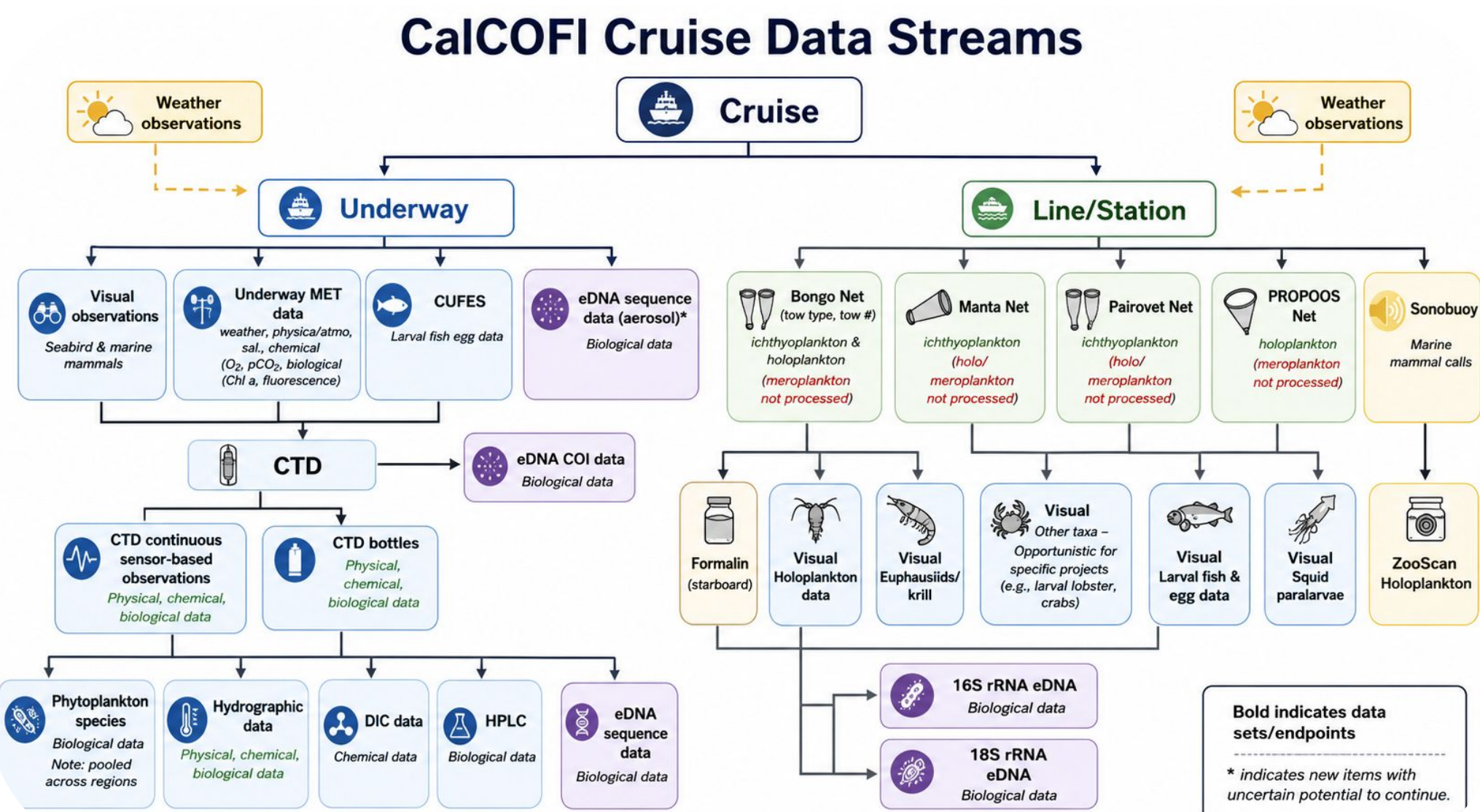
Datasets/Data Coverage

Data Coverage

- 75 / 113 stations
- Collection since 1950

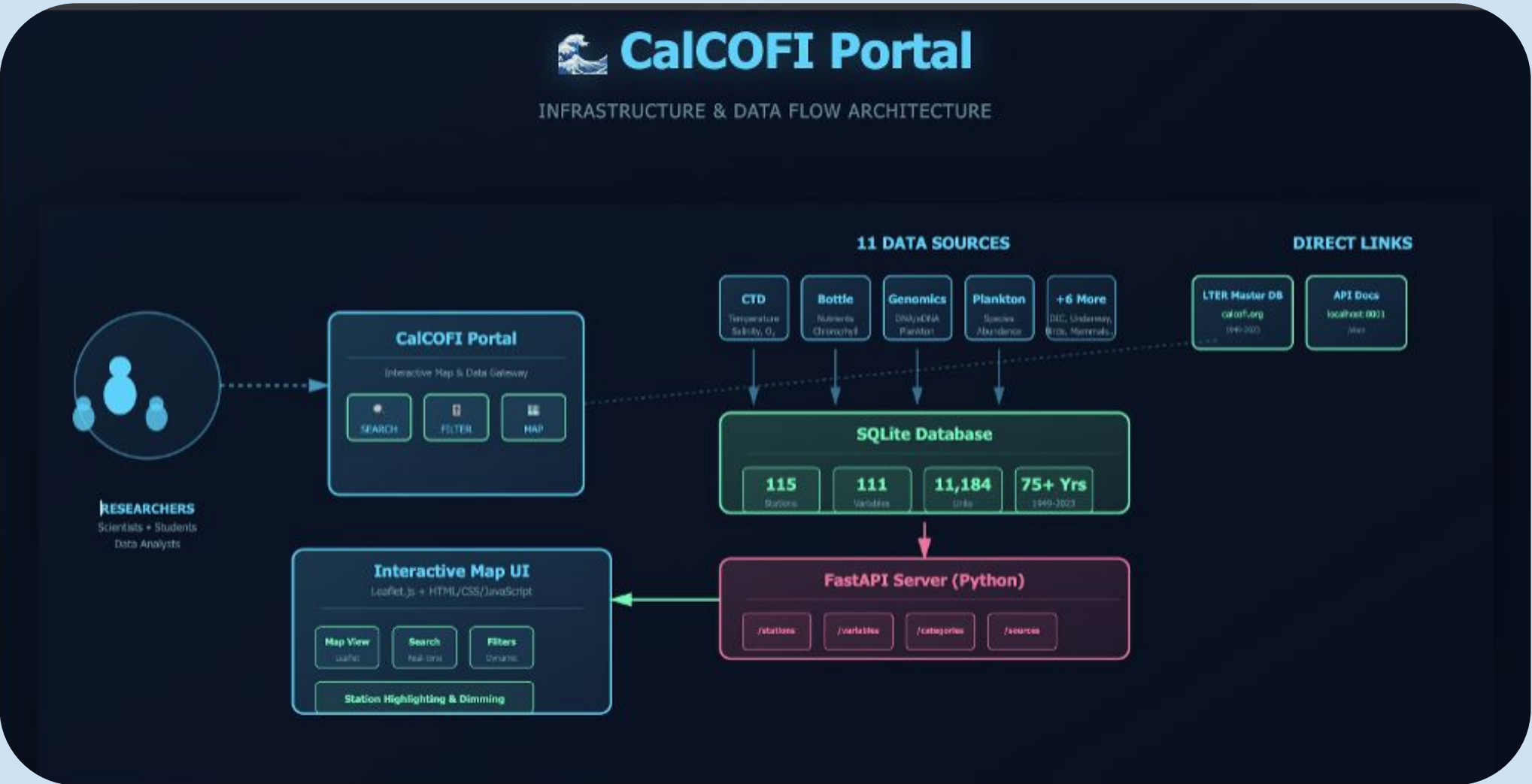
Data Sampling

- **CTD / Bottle**
 - Conductivity, Temperature, Depth rosette for real time analysis, equipped with Niskin bottles to capture samples for later laboratory analysis
- **Net Tows**
 - Data collected in transit between stations – thermosalinograph, meteorological & ship data collected by shipboard computer systems
- **Underway**
 - Bongo nets (paired 505 µm mesh nets with 71 cm diameter openings) towed obliquely through water columns to collect marine samples



Methodology / Portal Frameworks

- **Integrated** metadata from **multiple** CalCOFI datasets and repositories.
- **Mapped** variables and station records to source datasets and download links.
- Developed a **FastAPI backend** to support search, filtering, and data access workflows.
- Designed an intuitive **interface** for exploring stations, **variables**, and **data availability**.
- Structured the portal for **scalable deployment** and **future dataset expansion**.



Key Features

Metadata and Accessibility Options



Search/Sort Function (Year, Station, Variable)

